

ANURAN CALL SURVEYS ON SMALL WETLANDS IN PRINCE EDWARD ISLAND, CANADA RESTORED BY DREDGING OF SEDIMENTS

Authors: Stevens, Cameron E., Diamond, Antony W., and Shane Gabor, T.

Source: Wetlands, 22(1) : 90-99

Published By: The Society of Wetland Scientists

URL: [https://doi.org/10.1672/0277-5212\(2002\)022\[0090:ACSOSW\]2.0.CO;2](https://doi.org/10.1672/0277-5212(2002)022[0090:ACSOSW]2.0.CO;2)

BioOne Complete (complete.BioOne.org) is a full-text database of 200 subscribed and open-access titles in the biological, ecological, and environmental sciences published by nonprofit societies, associations, museums, institutions, and presses.

Your use of this PDF, the BioOne Complete website, and all posted and associated content indicates your acceptance of BioOne's Terms of Use, available at www.bioone.org/terms-of-use.

Usage of BioOne Complete content is strictly limited to personal, educational, and non - commercial use. Commercial inquiries or rights and permissions requests should be directed to the individual publisher as copyright holder.

BioOne sees sustainable scholarly publishing as an inherently collaborative enterprise connecting authors, nonprofit publishers, academic institutions, research libraries, and research funders in the common goal of maximizing access to critical research.

ANURAN CALL SURVEYS ON SMALL WETLANDS IN PRINCE EDWARD ISLAND, CANADA RESTORED BY DREDGING OF SEDIMENTS

Cameron E. Stevens¹, Antony W. Diamond², and T. Shane Gabor³

¹*Department of Biological Sciences*

University of Alberta

Edmonton Alberta, Canada T6G 2E1

E-mail: stevens@ualberta.ca

²*Atlantic Cooperative Wildlife Ecology Research Network*

University of New Brunswick

Fredericton New Brunswick, Canada E3B 6E2

³*Institute for Wetlands and Waterfowl Research*

Ducks Unlimited Canada

P.O. Box 1160

Stonewall Manitoba, Canada R0C 2Z0

Abstract: In 1990, the North American Waterfowl Management Plan (NAWMP) implemented a small-wetland restoration program in Prince Edward Island (PEI), Canada. Wetlands were restored by means of dredging accumulated sediment and organic debris to create open water and emulate pre-disturbance conditions. Three call surveys were conducted in the spring and summer of 1998 and 1999 to estimate abundance and occurrence of spring peepers (*Pseudacris crucifer*), wood frogs (*Rana sylvatica*), northern leopard frogs (*Rana pipiens*), American toads (*Bufo americanus*), and green frogs (*Rana clamitans*) on restored and reference wetlands. Numbers of species calling and abundance indices of northern leopard frogs, green frogs, and spring peepers were significantly higher on restored versus reference wetlands. The number of species calling in restored wetlands was positively correlated with proximity to freshwater rivers; in reference wetlands, the number was positively associated with forested perimeters and area of open water. Occurrence of calling green frogs in restored wetlands was positively correlated with percent cattail and, in reference wetlands, with proximity to other wetlands. Our results suggest that small wetland restoration projects may be a good conservation tool for anurans. We recommend further research on reproductive success and on local population trends in restored wetlands to determine if restoration is beneficial for anurans on PEI.

Key Words: wetland restoration, small wetland, NAWMP, anuran call survey, Marsh Monitoring Program

INTRODUCTION

Over the past 300 years, there has been a loss of wetlands, particularly small wetlands, in Prince Edward Island (PEI), Canada, by draining or filling for the purpose of “reclaiming” land for agriculture or settlement (Erskine 1992, R. Dibblee, provincial biologist, pers. comm.). In addition, potato farming has become an important industry, and the area of land planted with potatoes has increased steadily in recent years: 18 785 ha in 1959, 27 500 ha in 1989, and 46 500 ha in 1999 (Anon. 1999).

Low-lying wetlands can accumulate nutrient and sediment run-off from nearby agricultural fields, which contributes to accelerated rates of wetland degradation. PEI soils contain large amounts of silt and fine sand,

which result in weak aggregation or poor soil structure (Anon. 1999). Furthermore, the cultivation of steep slopes, row crops, and other exploitative agricultural practices result in accelerated soil erosion and accumulation of sediment in nearby aquatic ecosystems. Soil losses on agricultural fields average 20 tons/ha/yr and have been identified as the most important environmental problem in the province (Anon. 1999). Sedimentation and nutrient loading have been especially detrimental to small wetlands, which are important breeding habitats for some species of amphibians (Gibbs 1993, Semlitsch and Bodie 1998).

Ecosystem restoration is growing rapidly as a conservation practice but often without reliable understanding of the community response (Keddy 1999). The North American Waterfowl Management Plan

(NAWMP) has implemented a small wetland restoration program in PEI to enhance the landscape for waterfowl (Stevens *et al.* 2003). Since 1991, more than 100 small wetlands have been restored by dredging accumulated sediment. The recent concern over declining amphibian populations in various regions and habitats throughout the world (e.g., Barinaga 1990, Wake 1991) has drawn attention to conserving and restoring their habitats. Unfortunately, basic biological information on anuran habitat use and abundance is lacking in Canada (Green 1997).

Previous studies have documented amphibians exploiting a variety of man-made ponds, including bomb-craters (Warwick 1949), garden ponds (Beebe 1979, Banks and Laverick 1986), ponds created from timber harvesting (Jeffries 1991, McAlpine 1997), quarry ponds and roadside ditches (Cook 1967, McAlpine 1991), and ponds created on farmland (Baker and Halliday 1999). Therefore, dredging of soil and organic debris to restore small wetlands in PEI may affect local amphibian populations.

Our main objective was to compare use of restored and reference wetlands by anurans in PEI. A secondary objective was to determine relationships between habitat variables and anurans on small wetlands.

METHODS

Study Area

Prince Edward Island lies northeast of New Brunswick, between the Gulf of St. Lawrence and Northumberland Strait in eastern Canada (46°–47°N latitude). PEI has a gently rolling landscape, and the majority of land surface is less than 45 m above sea level. The crescent-shaped island has a deeply indented shoreline and is 225-km long and up to 56-km wide. About 4% (23 354 ha) of the island is freshwater wetland, and 1% (6 243 ha) is salt marsh (R. Dibblee, provincial biologist, pers. comm.).

The soils of PEI are primarily sandy loam, relatively shallow, acidic, and low in organic matter (Anon. 1999). The forests of Prince Edward Island are classified as a Sedimentary Lowland Mixed Forest and belong to the Acadian Forest Region (Rowe 1972). By the latter part of the 1800s, all of the original forest was cleared, and since then, white spruce (*Picea glauca* (Moench) Voss) and balsam fir (*Abies balsamea* (L.) P. Mill) have become the dominant woodland tree species (Anon. 1992). During the 1990s, approximately 50% of PEI was forested, and 40% was under agricultural use (Anon. 1992).

The island's climate is strongly influenced by the nearby ocean and south-flowing currents. Compared to the mid-continent, the arrival of spring is often late,

and minimum daily temperatures do not exceed zero consistently prior to mid-May (Anon. 1998). The climate is generally cool in summer; average temperatures in the Charlottetown area in April, May, June, and July are 2.3, 8.8, 14.4, and 18.4° C, respectively. The Maritimes are windier than most areas of Canada, and in PEI, winds average 20, 19, 17, and 16 km/h for April, May, June, and July, respectively. Precipitation averages 1050 mm annually and is distributed evenly throughout the year (Anon. 1998).

Study Sites

Our study included small freshwater wetlands (approximately 0.3 to 6 ha). Restored wetlands were dredged two to seven years prior to surveys (Table 1). The proportion of each restored wetland dredged varied from 30 to 95%, and the amount of dredged material left within a wetland (i.e., spoil) also differed among study wetlands. Study sites extended across the province from Malpeque Bay to Murray Harbour; most were concentrated near Charlottetown. Using information from a 1990 pre-restoration wetland inventory, we chose reference (i.e., control) wetlands within the range of pre-restoration characteristics of the dredged study sites. There were 46 individual study sites, of which only 20 were surveyed during both years due to logistical constraints. Fifteen restored and 15 reference wetlands were studied in 1998, and 18 (11 from 1998) restored wetlands and 18 (9 from 1998) reference wetlands in 1999.

We used water saturation and vegetation type to define the perimeter or boundary of all wetlands (Cowardin *et al.* 1979). Study sites were non-tidal, palustrine wetlands (i.e., basin wetlands) dominated by trees, shrubs, persistent emergents, emergent mosses, and lichens (Cowardin *et al.* 1979). The study wetlands formed either completely isolated natural wetland basins or were connected to a low order stream. Some wetlands were above a salt marsh and were exposed to brackish waters. Parts of these wetlands contained ocean-derived salts close to or in excess of 0.5%; however, the majority of cover was emergent and shrub vegetation associated with freshwater areas.

Dominant vegetation and water permanence varied among study sites. Emergent wetlands, by definition, are characterized by erect, rooted, herbaceous hydrophytes that are present for most of the growing season (Cowardin *et al.* 1979). The palustrine emergent wetland sites were dominated primarily by cattails (*Typha latifolia* L.), bulrushes (*Scirpus* spp.), rushes (*Juncus* spp.), and sedges (*Carex* spp.). Wetlands above an estuary with both palustrine and estuarine characteristics contained cattails (*T. latifolia* and *T. angustifolia* L.), bulrush (*Scirpus pungens* Vahl), arrow grass (*Triglo-*

Table 1. Mean ($\pm$ SE) landscape- and wetland-level characteristics of 22 restored (i.e., dredged) and 24 reference wetlands studied in PEI during 1998 and 1999. Wetland vegetation and water measurements reflect post-restoration conditions.

	Habitat Variable	Restored (n = 22)	Reference (n = 24)
Restoration variables*	Years since restoration	3.36 (0.35)	n.a.
	Percent dredged	68.0 (3.28)	n.a.
	Area dredged (ha)	1.42 (0.16)	n.a.
Wetland size	Area (ha)	2.3 (0.34)	2.08 (0.28)
Water measurements	Open water (%)	49.0 (4.1)	4.5 (1.4)
	Open water (ha)	1.02 (0.13)	0.10 (0.04)
	Water depth (cm)	69.2 (3.65)	19.6 (4.47)
Vegetation type	Cattail (%)	13.9 (2.2)	24.7 (3.9)
	Shrub (%)	13.3 (2.4)	33.2 (6.41)
	Meadow (%)	23.4 (4.4)	32.9 (6.5)
	Peatland (%)	1.9 (0.7)	5.8 (2.01)
	Cattail (ha)	0.36 (0.09)	0.52 (0.11)
	Shrub (ha)	0.42 (0.12)	0.77 (0.19)
	Meadow (ha)	0.49 (0.09)	0.58 (0.15)
	Peatland (ha)	0.06 (0.03)	0.14 (0.06)
Perimeter type (%)	Forest	35.2 (8.0)	34.6 (5.8)
	Agriculture	47.5 (6.9)	41.5 (7.6)
	Salt marsh	6.1 (2.8)	6.7 (3.4)
	Road	4.8 (1.9)	12.3 (3.1)
	Grassland	6.4 (2.1)	4.8 (1.3)
Nearest landscape feature (km)	Coast	6.82 (1.73)	5.54 (0.92)
	Estuary or large tidal river	2.44 (0.6)	2.34 (0.5)
	Fresh water pond	1.76 (0.29)	1.51 (0.25)
	Large-order river	0.80 (0.21)	0.59 (0.15)
	Deep emergent marsh	0.81 (0.15)	0.81 (0.19)
	Wetland of similar size	0.86 (0.19)	0.74 (0.28)
	3 wetlands of similar size	1.58 (0.47)	0.98 (0.29)
	River (any size)	0.16 (0.07)	0.06 (0.03)
	Salt marsh	2.57 (0.62)	3.02 (0.63)

* Variables not used in PCA and stepwise regression.

chin maritima L.), and reed bent grass (*Calamagrostis stricta* (Timm) Koel.). Study sites with a temporary water regime were dominated by grasses, such as blue-joint grass (*C. canadensis* (Michx.) Beauv.) and reed-canary grass (*Phalaris arundinacea* L.), sedges, rushes, goldenrods (*Solidago* spp.), asters (*Aster* spp.), and bulrushes (*S. cyperinus* (L.) Kunth and *S. atrocinctus* Fern.). Scrub-shrub wetlands include wetlands dominated by woody vegetation less than 6m tall (Cowardin et al. 1979). The dominant species among such study sites included alders (*Alnus incana* L.), willows (*Salix* spp.), sweet gale (*Myrica gale* L.), and red maple (*Acer rubrum* L.).

Open water created from wetland dredging provided habitat for flora such as cattail (*Typha* spp.), common duckweed (*Lemna minor* L.), water-starwort (*Callitriche* spp.), water lilies (*Nuphar luteum* (L.) Sm. and *Nymphaea odorata* Ait.), pondweeds (*Potamogeton* spp.), macroalgae (*Chara* spp.), and northern wild rice (*Zizania palustris* L.). The dredged material supported species such as goldenrods (*Solidago* spp.), asters,

grey birch (*Betula populifera* Marsh.), raspberry (*Rubus* spp.), spotted-touch-me-not (*Impatiens capensis* Meerb.), bedstraw (*Galium* spp.), clover (*Trifolium* spp.), cinquefoil (*Potentilla* spp.), and grasses (Poaceae family).

Surveys

Three call surveys (Shirose et al. 1997) were conducted during the spring and early summer of 1998 and 1999 to coincide with the optimum breeding period of frogs and toads found on the island. The first survey (3–15 May 1998, 1–5 May 1999) targeted early breeders like the spring peeper (*Pseudacris crucifer* (Wied-Neuweid)) and wood frog (*Rana sylvatica* Le Conte); the second survey (17–24 June 1998, 25–31 May 1999) coincided with mid-season breeding anurans like the northern leopard frog (*Rana pipiens* Schreber), and American toad (*Bufo americanus* Holbrook); and the last survey (July 13–20, 1998; June 22–26, 1999) targeted the late-breeding green frog

(*Rana clamitans* Latreille). The surveys followed methods of the Marsh Monitoring Program (Anon. 1997). Observers estimated the relative abundance of species calling by ranking the density of call clusters as 0 = no calls heard, 1 = individual calls not overlapping, 2 = some overlapping calls, but the number of individual calling can be reliably estimated, or 3 = a continuous chorus of calls in which individual calls could not be identified. The number of individuals within a group was counted if a rank of 1 or 2 was recorded. To avoid observer bias, we spent two nights in the field prior to each survey to ensure observer agreement with the recording of each species and their relative abundance. The observers were S. Gaudet (during 1998), J. Nicholson (during 1999), and C. E. Stevens (during 1998 and 1999).

Surveys were completed during evenings with ideal weather conditions (e.g., high humidity, weak winds, and warm temperatures). We recorded wind strength using the Beaufort Scale (World Meteorological Organization 1970), air temperature, survey time, percent cloud cover and the presence of rain (0 = dry, 1 = damp/fog, 2 = drizzle, and 3 = rain) during each survey.

A 180° point count was used to estimate the species composition and abundance of calling anurans on each site. Each wetland had one or two survey stations, depending on the size of the wetland. Large sites had two survey stations with point counts made from the center of the wetland, back-to-back, so each survey station faced in opposite directions. The small size of the study sites (<6 ha) and back-to-back survey stations ensured that wetland habitat being surveyed extended no more than 150 m from a point count. To allow nearby frogs to acclimatize to our presence after we approached a survey station, we allowed one minute to pass before recording began. Surveys were continued for three minutes. Longer surveys rarely result in new species being identified (Shirose *et al.* 1997).

Habitat characteristics (Table 1) were quantified visually in mid-summer during both survey years. Aerial photographs were taken in 1990, and a planimeter was used to determine pre-management wetland characteristics and wetland area. During the first week of May, water-depth measurements were recorded on 50-m random transects in the center of each wetland (every 10 m). Topographic maps (1:50 000) and the provincial GIS database were used to calculate landscape metrics (e.g., distance to freshwater stream; Table 1).

Statistics

Statistical analyses of abundance indices determined from anuran call surveys were based on the highest rank recorded for each species per site. Number of

species calling and abundance indices were compared between years and between restored and reference wetlands using a linear mixed effects model ANOVA (Zar 1999) in SPLUS 2000. Mixed effects models are a flexible tool for analyzing grouped data and incorporate both fixed and random effects. The model groups data by associating common random effects to observations sharing the same level of a classification factor. Thus, the model integrated our study design of some wetlands surveyed over both years and other surveyed during one year only (1998 or 1999). Data on number of species calling were square root- or log-transformed where necessary to satisfy assumptions of parametric procedures.

Anuran abundance indices are ordinal data and were thus subjected to non-parametric techniques. Non-parametric procedures can be viewed as a parametric analysis of ranked data (Conover and Iman 1981). Thus, we rank-transformed the response variable (i.e., frog call index of abundance) prior to testing for differences between wetland type and year.

To determine anuran-habitat relationships, we used Principal Components Analysis (PCA; based on the correlation matrix), stepwise linear regression, and logistic regression. PCA simplified the dataset by reducing the number of environmental variables to a smaller number of orthogonal components that maximized the variance accounted for in the data (Rotenberry and Wiens 1980). We treated an environmental component as relevant if it had an eigenvalue greater than 1.0. We used stepwise regression of the environmental components on each response variable. Data from sites covered during both 1998 and 1999 were pooled (i.e., averaged and represented by $n = 1$) for the northern leopard frog, green frog, and American toad regression models. Analyses of residuals were performed to ensure that assumptions of parametric analyses were met.

The first round of surveys in 1998 extended too long (12 days) to accurately estimate wood frog abundance and occurrence. Table 2 shows higher wood frog abundance indices in 1999 versus 1998 ($df = 1, 19$, $F = 43.95$, $p < 0.01$). Therefore, wood frog data collected in 1998 were excluded from regression models and ANOVA tests. Furthermore, wood frogs were also excluded from estimates on the number of species calling per wetland. All significance levels were set at 0.10 (a priori), which increased the probability of a Type I error (rejecting true null hypothesis) and decreased the probability of a Type 2 error (accepting false null hypothesis).

RESULTS

The five species of anurans found on PEI (wood frogs, spring peepers, American toads, northern leopard

Table 2. Mean ($\pm$ SD) index of abundance for anuran species and number of species calling for restored and reference wetlands and during 1998 and 1999. P refers to differences between the two preceding cells according to an LME model ANOVA.

Species	Restored (n = 22)*	Reference (n = 24)*	P	1998 (n = 20)	1999 (n = 20)	p
Wood frog	1.45 (0.26)	1.17 (0.23)	n.s.	0.55 (0.15)	2.05 (0.27)	<0.01
Wood frog**	2.17 (0.28)	1.89 (0.32)	n.s.	—	—	—
Spring peeper	2.68 (0.12)	2.10 (0.23)	0.043	2.7 (0.16)	2.15 (0.28)	0.034
Am. toad	1.5 (0.24)	1.08 (0.22)	n.s.	1.5 (0.27)	1.1 (0.25)	n.s.
Leopard frog	0.5 (0.10)	0.23 (0.11)	0.037	0.5 (0.14)	0.55 (0.15)	n.s.
Green frog	0.68 (0.15)	0.21 (0.08)	0.013	0.55 (0.15)	0.55 (0.14)	n.s.
# of species	2.68 (0.17)	1.77 (0.22)	<0.01	2.6 (0.24)	2.4 (0.30)	n.s.

n.s. = non-significant.

* Sites covered over both years were pooled for presentation in table.

** 1999 data only and P-value determined from a one-way ANOVA.

ard frogs, and green frogs) were present on both restored and reference wetlands in 1998 and 1999. The number of species detected by our call surveys was higher on restored versus reference wetlands ($df = 1,44$, $F = 9.72$, $p < 0.01$); number of species calling was not significantly different between years ($df = 1,19$, $F = 1.88$, $p = 0.19$).

Spring peeper abundance indices were significantly higher on restored versus reference wetlands ($df = 1,44$, $F = 4.3$, $p = 0.043$). The indices also varied significantly between years ($df = 1,19$, $F = 5.2$, $p = 0.034$); peepers were more abundant in 1998. Wood frog abundance indices did not differ between restored and reference wetlands using 1998 and 1999 data ($df = 1,44$, $F = 0.4$, $p = 0.53$) or 1999 data alone ($df = 1,34$, $F = 0.3$, $p = 0.59$). Northern leopard frog ($df = 1,44$, $F = 4.62$, $p = 0.037$) and green frog ($df = 1,44$, $F = 6.75$, $p = 0.013$) abundance indices were significantly greater on restored versus reference wetlands. There were no significant differences in abundance indices between years for the northern leopard frog ($df = 1,19$, $F = 0.02$, $p = 0.88$) or green frog ($df = 1,19$, $F < 0.01$, $p = 0.95$). American toad abundance indices were not significantly different between years ($df = 1,19$, $F = 2.05$, $p = 0.17$) or between restored and reference wetlands ($df = 1,44$, $F = 1.27$, $p = 0.27$).

A PCA (based on the correlation matrix) on environmental variables resulted in 8 components that explained 79.8% of the total variation (Table 3). Component 1 represents a gradient from shrub habitat and a forested perimeter (high loadings) to meadow vegetation and an agricultural perimeter (low loadings). High loadings on component 2 represent wetlands in close proximity to other wetland habitat. Greater water depths and amount of open water loaded high on component 3. High loadings on component 4 refer to large wetlands with greater area of cattail. Component 5 represents a gradient of wetlands near a tidal river or salt marsh (high loadings) to wetlands near a freshwater

river or stream (low loadings). High loadings on component 6 represent wetlands with greater amounts of peatland habitat. On component 7, percent cattail cover scored low, whereas proximity to a pond or lake scored high. On component 8, low loadings represent wetlands in close proximity to coast and high loadings represent proximity to a large freshwater river.

To identify significant correlations of habitat variables with anurans in restored and reference wetlands, we conducted stepwise regression using the eight principal components of the habitat data against the number of species calling and occurrence of calling wood frogs, spring peepers, American toads, northern leopard frogs, and green frogs. Results are summarized in Table 4.

In restored wetlands, the number of species calling was positively correlated with close proximity to freshwater rivers (components 5 and 8). Calling green frog occurrence was positively correlated with percent cattail (component 7). Some species were ubiquitous on restored wetlands (i.e., wood frog (83%) and spring peeper (100%)), and no species occurrence-habitat relationships were noted.

In reference wetlands, the number of species calling was positively correlated with forested perimeters and shrub ecotones (component 1), and greater water depths and open water (component 3). Calling green frog occurrence was positively associated with close proximity to other wetland habitat (component 2). Calling wood frog occurrence was positively correlated with greater water depths and open water (component 3). Calling American toad presence was positively correlated with wetland area (component 4) and peatland habitat (component 6). Significant species occurrence-habitat relationships were not detected for spring peepers, which were present on almost all (88%) of the reference sites, nor were relationships detected for northern leopard frogs, which were present on relatively few (17%) sites.

Table 3. Principal components of environmental data determined by a PCA (based on the correlation matrix) of 24 reference wetlands and 22 restored wetlands. Only those loadings with values $>|0.55|$ are included in table and interpreted in results.

Variable	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8
Eigenvalue	3.67	3.36	3.02	2.79	2.58	2.25	1.60	1.46
Variance explained	14.13	12.93	11.62	10.72	9.94	8.67	6.17	5.62
Cumulative variance	14.13	27.07	38.69	49.40	59.35	68.02	74.19	79.81
Wetland area (ha)	—	—	—	0.830	—	—	—	—
Water depth	—	—	0.922	—	—	—	—	—
Cover Type								
Open water (%)	—	—	0.917	—	—	—	—	—
Open water (area)	—	—	0.831	—	—	—	—	—
Cattail (%)	—	—	—	—	—	—	−0.687	—
Cattail (area)	—	—	—	0.859	—	—	—	—
Shrub (%)	0.746	—	—	—	—	—	—	—
Shrub (area)	0.666	—	—	—	—	—	—	—
Meadow (%)	−0.692	—	—	—	—	—	—	—
Meadow (area)	—	—	—	—	—	—	—	—
Peatland (%)	—	—	—	—	—	0.879	—	—
Peatland (area)	—	—	—	—	—	0.925	—	—
Perimeter Type								
Forested	0.784	—	—	—	—	—	—	—
Agriculture	−0.895	—	—	—	—	—	—	—
Salt marsh	—	—	—	—	0.768	—	—	—
Road	—	—	—	—	—	—	—	—
Grassland	—	—	—	—	—	—	—	—
Proximity to Landscape Feature*								
Coast	—	—	—	—	—	—	—	−0.582
Estuary or large tidal river	—	—	—	—	0.809	—	—	—
Pond or lake	—	—	—	—	—	—	0.675	—
Large fresh-water river	—	—	—	—	—	—	—	0.831
Deep emergent marsh	—	0.850	—	—	—	—	—	—
Wetland of similar size	—	0.872	—	—	—	—	—	—
3 wetlands of similar size	—	0.842	—	—	—	—	—	—
Fresh-water river (any size)	—	—	—	—	−0.607	—	—	—
Salt marsh	—	—	—	—	0.629	—	—	—

* Transformations of landscape measurements ($1/(\text{variable} + 0.01)$) were made prior to PCA.

DISCUSSION

Anuran Call Surveys

Long-term monitoring programs based on amphibian call count surveys have been implemented in various regions of North America (e.g., North American Amphibian Monitoring Program (Anon. 1995), Marsh Monitoring Program (Anon. 1997)). In eastern Canada, programs such as the Marsh Monitoring Program (MMP) have been restricted primarily to the Great Lakes Region, where survey schedules are based on latitude and air temperatures. For centrally located areas (43rd to 47th parallel), the MMP recommends a starting date between April 15 and 30 and a first survey minimum air temperature of 8° C for anuran call surveys. Prince Edward Island lies on the 46th parallel, and our 1998 and 1999 survey schedule for American

toads, leopard frogs, and green frogs differed by a month due to variation in the onset of calling and local environmental variables. The climate of PEI is strongly influenced by the Atlantic Ocean, with minimum daily air temperatures rarely exceeding 8° C during April, May, and early June (Anon. 1998). Unpredictable weather, late spring temperatures, and high winds of the Maritimes suggest a very narrow window of opportunity for breeding anurans and significant annual variations in the onset of reproduction.

Yearly differences in our wood frog abundance indices may reflect narrow breeding windows, yearly variations in breeding times, and an inappropriate survey schedule. Wood frogs are explosive breeders, and in 1998, our survey for this species extended for a long period (12 days). Wood frogs were recorded at the start of the survey period, but few were noted toward

Table 4. Results of stepwise linear and logistic regression on number of anuran species calling and species calling occurrence in reference and restored wetlands. Variables are presented in the order they were entered into the model. All models have a $P < 0.05$.

	Year	R ²	df	Variable	Estimate	t	P
Reference							
# of species	'99	0.51	2,15	Intercept	2.04	13.34	<0.01
				PC 1	0.24	2.54	0.023
				PC 3	0.57	3.30	<0.01
Wood frog	'99	0.39*	1,16	Constant	9.22	1.68	0.094
				PC 3	8.67	1.66	0.097
American toad	'98–99	0.54*	4,19	Constant	4.06	1.66	0.096
				PC 2	−1.53	−1.47	n.s.
				PC 4	6.75	1.76	0.078
				PC 5	5.91	1.62	n.s.
				PC 6	4.04	1.82	0.068
Green frog	'98–99	0.23*	1,22	Constant	−1.96	−2.65	<0.01
				PC 2	1.13	2.08	0.038
Restored							
# of species	'99	0.42	3,14	Intercept	1.98	−2.56	<0.01
				PC 5	−0.14	−2.56	0.023
				PC 7	−0.19	−1.63	n.s.
				PC 8	0.15	1.78	0.096
Green frog	'98–99	0.23*	1,20	Constant	0.78	1.4	n.s.
				PC 7	−2.26	−2.18	0.029

n.s. = non-significant.

* R² refers to McFadden's Rho² and T based on likelihood ratio (Kleinbaum 1994).

the end of the survey period. A shorter survey schedule focused on local climate patterns would have improved species detectability. Survey schedules should also incorporate local water-temperature measurements and humidity indices, as they seem to influence the start and end of breeding for anurans (Green 1997, Lepage et al. 1997). However, despite being confident that our sampling regime was well within the breeding (or calling) window for the spring peeper, we still observed yearly differences in abundance for this species. Such yearly differences may reflect natural fluctuations in recruitment and population size typical of most amphibian species (e.g., Pechmann et al. 1991).

Restoration and Habitat Use

Restored and reference wetlands were used by all species of anurans found on PEI. Similarly, previous research has found that man-made waterbodies can support various species of amphibians (Cook 1967, Cooke 1975, Beebee 1979, 1985, Banks and Laverick 1986, McAlpine 1991, 1997). Restored wetlands had higher numbers of anuran species calling than did reference wetlands. In addition, based on indices determined from call surveys, spring peepers, green frogs, and northern leopard frogs were more abundant in restored versus reference wetlands.

Wetland dredging created semi-permanent water

bodies within a previously closed or vegetation-choked wetland. Reference study sites typically had shallow water depths (0.2 m) and little open water (5%), and many were dry by mid-June. As was expected, water depths and open water were positively correlated with number of species calling in reference wetlands. An extended hydroperiod in reference wetlands or an increase in hydroperiod created by wetland dredging reduced the possibility of ponds drying and tadpoles dying in natal ponds prior to metamorphosis and emigration to terrestrial habitats (e.g., Pechmann et al. 1989).

Unpredictably, the only anuran in reference wetlands correlated with an increase in hydroperiod was the wood frog, a species typically associated with temporary or ephemeral ponds (Cook 1984, Gilhen 1984). This relationship raises the question of why we did not detect higher wood frog abundance indices on restored versus reference wetlands. Perhaps, detrimental effects associated with restored wetlands offset the benefits of increased open water for wood frogs. Such detrimental effects could include an increase of fish and potential predation in restored wetlands (e.g., Hecnar and M'Closkey 1997a).

Increases in open water and water depths due to restoration also provided habitat for green frogs and northern leopard frogs. Green frogs breed in semi-permanent water bodies in early to mid-summer when

ephemeral or temporary ponds are dry (Cook 1984, Gilhen 1984). Furthermore, deeper water may not freeze to the bottom, providing winter hibernacula for species such as the green frog and northern leopard frog (Cook 1984, Cunjak 1986). A positive northern leopard frog response to wetland restoration could also reflect the presence of spoil piles created during wetland dredging. The spoil piles may have provided the meadow-like habitat preferred by post-breeding northern leopard frogs (Cook 1984, Gilhen 1984, McAlpine 1997).

Presence of a forested perimeter, shrub habitat, and open water is representative of undisturbed natural emergent wetlands in the region, and these characteristics were positively correlated with number of species calling in reference wetlands. A forested perimeter and shrub ecotone may provide refugia or movement corridors for post-breeding and newly metamorphosed individuals (e.g., wood frogs; deMaynadier and Hunter 1999). A forested perimeter may also serve as a buffer from agricultural activities. Agricultural activities, such as pesticide use, have been linked to larval death and limb deformities in amphibians (e.g., Berrill *et al.* 1997, Ouellet *et al.* 1997), and cattle grazing has been linked to smaller populations of wood frogs and chorus frogs (*Pseudacris maculata* (Agassiz)) (Ambrose and Paszkowski 1998). Nutrient run-off and sedimentation associated with agricultural landscapes are also detrimental to anurans (Corn and Bury 1989, Hecnar 1995, Perison *et al.* 1997; Bishop *et al.* 1999).

American toad occurrence in reference wetlands was correlated with large wetlands with greater cattail area. Large wetlands may include a diversity of microhabitats that play a key role in anuran distribution (McAlpine and Dilworth 1989). Extensive floating mats of dead cattail noted in large wetlands can provide ideal calling posts for male American toads during the breeding season (D. McAlpine, New Brunswick Museum, pers. comm.). Although American toads are capable of using a wide variety of wetland habitats (Cook 1967, Gilhen 1984), such floating mats were typically absent in restored wetlands, which may explain the relatively lower American toad abundance indices on restored wetlands.

American toad occurrence in reference wetlands was also positively correlated with peatland habitat. Peatland habitat was typical in areas of low agricultural land use (C. E. Stevens, pers. obs.) and, thus, may represent better-quality breeding habitat. Nutrient run-off from agriculture can be a contributing factor to lower reproductive success and density of American toads (Bishop *et al.* 1999).

Anuran species found on PEI breed in ponds and move to terrestrial or other aquatic environments to forage or overwinter (Cook 1964, 1987). Individuals

often return to their natal pond to breed in subsequent years, although a small percentage disperses to other ponds (e.g., wood frog; Berven and Grudzien 1990). The long-term persistence of amphibian metapopulations is dependent on successful dispersal movements and recolonization of wetlands (Gill 1978, Sjögren-Gulve 1994). Like our study, previous research has documented that close proximity to aquatic habitat influences amphibian populations and species richness (Sjögren-Gulve 1994, Hecnar and M'Closkey 1997b, Semlitsch and Bodie 1998, Baker and Halliday 1999).

In restored wetlands, numbers of species calling were positively correlated with proximity to freshwater rivers and streams. Foraging or dispersal movements may be beneficial along freshwater rivers and associated riparian zones in PEI because they provide wet microhabitats and protection from unfavorable climatic conditions in an otherwise open landscape. Furthermore, freshwater wetlands in PEI are dispersed due to agricultural practices, thereby emphasizing the important role streams play in maintaining connectivity. Wetland isolation can make amphibian populations more susceptible to pollution, habitat destruction, or changes in water levels. Recognizing the importance of streams in maintaining connectivity and population persistence of species is necessary if ecosystem integrity and overall biodiversity are management goals.

In reference wetlands, green frog occurrence was associated with close proximity to other wetland habitats. Reference wetlands typically lacked open water but, in some cases, were in close proximity to other wetlands with open water. Adjacent open water areas may have been the source (e.g., Pulliam 1988) for individuals noted in the reference wetlands. In contrast, green frog occurrence in restored wetlands was linked to local habitat characteristics (i.e., percent cattail). Cattail may have served as refugia from predators (e.g., Babbitt and Tanner 1998, Andersen *et al.* 1999).

In contrast to the green frog, the American toad, spring peeper, and wood frog are less dependent on permanent water sources (Gilhen 1984, Cook 1984) and were not correlated with any of our measured landscape variables. In addition, these species were encountered calling in a variety of habitats, including freshwater wetlands and slightly brackish wetlands adjacent to an estuary. Because of the numerous tidal rivers and highly indented shoreline in PEI, species such as the American toad may benefit due to their ability to exploit both freshwater and brackish marsh habitats (Kiviat and Stapleton 1983). Other anurans, such as the northern leopard frog, have been documented in brackish habitats of Lower Hudson Bay (Klemens *et al.* 1987); however, the ability of the wood frog and spring peeper to exploit such habitats in PEI is unknown and warrants further study. It is

more likely that these species were using microhabitats dominated by freshwater within the brackish areas.

Limitations of Study

Landscape features, such as forest cover (Findlay and Houlihan 1996, Hecnar and M'Closkey 1998), agricultural activity (Ouellet et al. 1997, Bishop et al. 1999), presence of predatory fish (e.g., Hecnar and M'Closkey 1997a), or water chemistry (e.g., Hecnar and M'Closkey 1996, Andersen et al. 1999) were not accounted for in this study and may influence the abundance and composition of anurans in small wetlands. Furthermore, without information on reproductive success, we lack adequate information to verify the effects of habitat quality on anuran recruitment (Van Horne 1983). Some wetlands may have served as sinks (e.g., Pulliam 1988), and this would not have been detected. In addition, our units of measure for this study were based on call surveys alone. To estimate amphibian assemblages and abundance in wetlands reliably, additional survey techniques, such as pitfall trapping and visual searches, should accompany call surveys (Corn 1994, Crump and Scott 1994). However, despite the inherent biases and difficulties associated with call surveys, we believe that our results are particularly relevant due to the lack of baseline information on PEI amphibians and the sound framework they provide for further research.

Conclusion

Cook (1967) found amphibian populations to be abundant on PEI during the late 1950s despite significant changes to the landscape. He noted abundant wood frogs, spring peepers, and American toads and common-to-abundant populations of northern leopard frogs and green frogs. However, agricultural land use has intensified since the 1950s, which may threaten the viability of local amphibian populations. In 1998 and 1999, we noted greater use of restored versus reference wetlands by anurans in PEI. It seems that anurans are well-adapted at exploiting new habitats (e.g., Baker and Halliday 1999). We conclude that wetland restoration by means of dredging accumulated organic soil and debris for the primary purpose of creating open water for waterfowl does provide habitat for anurans.

ACKNOWLEDGMENTS

Financial support for this study was provided by Ducks Unlimited Canada (DUC) through the Institute for Wetland and Waterfowl Research (IWWR), the North American Wetlands Conservation Council (NAWCC), and Ducks Unlimited Inc. (DUI) through

the Eastern Habitat Joint Venture (EHJV), Wildlife Habitat Canada (WHC), Environment Canada (CWS), and Prince Edward Island—Technology and Environment. Field assistance was provided by Jane Nicholson and Scott Gaudet. Helpful advice on preparation of the manuscript was provided by Laurel Bernard, Don McAlpine, Henry Murkin, and one anonymous reviewer. Statistical advice was provided by Llewellyn Armstrong and Falk Huettmann. This is ACWERN publication no. UNB-23.

LITERATURE CITED

- Anonymous. 1992. 1990/92 Prince Edward Island Forest Inventory Summary. Silviculture and Development Section, Forestry Branch, Department of Energy and Forestry. Charlottetown, PEI, Canada.
- Anonymous. 1995. North American Amphibian Monitoring Program. U. S. Geological Survey, Patuxent Wildlife Research Center [on-line]. Available: <http://www.mp2-pwrc.usgs.gov/naamp/> [August 12, 2001].
- Anonymous. 1997. Marsh Monitoring Program Training Kit. Long Point Bird Observatory and Environment Canada. Great Lakes Protection Fund. Port Rowan, ON, Canada.
- Anonymous. 2001. The Green Lane. Environment Canada Atlantic Region [on-line]. available: <http://www.ns.ec.gc.ca/> [Nov. 15, 2001].
- Anonymous. 1999. PEI Department of Fisheries and Environment, PEI Department of Agriculture and Forestry, Environment Canada, and Agriculture and AgriFood Canada. The best management practices for soil Conservation for Potato Production. Charlottetown, PEI, Canada.
- Ambrose, N. E. and C. A. Paszkowski. 1998. The effect of timing of cattle grazing on amphibian abundance in central Alberta ponds. Canadian Association of Herpetology Bulletin 12:12–13.
- Anderson, A. M., D. A. Haukos, and J. T. Anderson. 1999. Habitat use by anurans emerging and breeding in playa wetlands. Wildlife Society Bulletin 27:759–769.
- Babbitt, K. J. and G. W. Tanner. 1998. Effects of cover and predator size on survival and development of *Rana utricularia* tadpoles. Oecologia 114:258–262.
- Baker, J. M. R. and T. R. Halliday. 1999. Amphibian colonization of new ponds in an agricultural landscape. Herpetological Journal 9:55–63.
- Banks, B. and G. L. Laverick. 1986. Garden ponds as amphibian breeding sites in a conurbation in the north east of England (Sunderland, Tyne and Wear). Herpetological Journal 1:44–50.
- Barinaga, M. 1990. Where have all the froggies gone? Science 147: 1033–34.
- Beebee, T. J. C. 1979. Habitats of the British amphibians: suburban parks and gardens. Biological Conservation 15:241–257.
- Beebee, T. J. C. 1985. Discriminant analysis of amphibian habitat determinants in South-East England. Amphibia-Reptilia 6:35–43.
- Berrill, M., S. Bertram, and B. Pauli. 1997. Effects of pesticides on amphibian embryos and larvae. Herpetological Conservation 1: 233–245.
- Berven, K. A. and T. A. Grudzien. 1990. Dispersal in the wood frog (*Rana sylvatica*): implications for genetic population structure. Evolution 44:2047–2056.
- Bishop, C. A., N. A. Mahony, J. Struger, P. NG, and K. E. Pettit. 1999. Anuran development, density, and diversity in relation to agricultural activity in the Holland River watershed, Ontario, Canada (1990–1992). Environmental Monitoring and Assessment 57: 21–43.
- Conover, W. J. and R. L. Iman. 1981. Rank transformations as a bridge between parametric and nonparametric statistics. The American Statistician 35:124–133.
- Cook, F. R. 1967. An analysis of the herpetofauna of Prince Edward

- Island. National Museum of Canada. Ottawa, ON, Canada. Bulletin No. 212, Biological Series No. 75.
- Cook, F. R. 1984. Introduction to Canadian Amphibians and Reptiles. National Museums of Canada. Ottawa, ON, Canada.
- Cooke, A. S. 1975. Spawn site selection and colony size of the frog (*Rana temporaria*) and the toad (*Bufo bufo*). *Journal of Zoology*, London 175:29–38.
- Corn, P. S. 1994. Straight-line drift fences and pitfall traps. p. 109–118. In W. A. Heyer, M. A. Donnelly, R. W. McDiarmid, L. C. Hayek, and M. S. Foster (eds.) *Measuring and Monitoring Biological Diversity: Standard Methods for Amphibians*. Smithsonian Institution Press, Washington, DC, USA.
- Corn, P. S. and R. B. Bury. 1989. Logging in western Oregon: responses of headwater habitats and stream amphibians. *Forest Ecology and Management* 29:39–57.
- Cowardin, L. M., V. Cater, F. C. Golet, and E. T. Laroe. 1979. Classification of wetlands and deepwater habitats of the United States. U. S. Fish and Wildlife Service, Washington, DC, USA. OBS-79/31.
- Crump, M. L. and N. J. Scott, Jr. 1994. Visual encounter surveys. p. 84–91. In W. A. Heyer, M. A. Donnelly, R. W. McDiarmid, L. C. Hayek, and M. S. Foster (eds.) *Measuring and Monitoring Biological Diversity: Standard Methods for Amphibians*. Smithsonian Institution Press, Washington, DC, USA.
- Cunjak, R. A. 1986. Winter habitat of northern leopard frogs, *Rana pipiens*, in a southern Ontario stream. *Canadian Journal of Zoology* 64:255–257.
- DeMaynadier, P. G. and M. L. Hunter, Jr. 1999. Forest canopy closure and juvenile emigration by pool-breeding amphibians in Maine. *Journal of Wildlife Management* 63:441–450.
- Erskine, A. J. 1992. Atlas of Breeding Birds of the Maritime Provinces. Nimbus Publications Ltd. and Nova Scotia Museum, Halifax, NS, Canada.
- Findlay, C. S. and J. Houlahan. 1996. Anthropogenic correlates of species richness in southeastern Ontario wetlands. *Conservation Biology* 11:1000–1009.
- Gibbs, J. P. 1993. Importance of small wetlands for the persistence of local populations of wetland-associated animals. *Wetlands* 13: 25–31.
- Gilhen, J. 1984. Amphibians and Reptiles of Nova Scotia. Nova Scotia Museum, Halifax, NS, Canada.
- Gill, D. E. 1978. The metapopulation ecology of the red-spotted newt, *Notophthalmus viridescens* (Rafinesque). *Ecological Monographs* 48:145–166.
- Green, D. M. 1997. Perspectives on amphibian population declines: Defining the problem and searching for answers. *Herpetological Conservation* 1:291–308.
- Hecnar, S. J. 1995. Acute and chronic toxicity of ammonium nitrate fertilizer to amphibians from southern Ontario. *Environmental and Toxicological Chemistry* 14:2131–2137.
- Hecnar, S. J. and R. T. M'Closkey. 1996. Amphibian species richness and distribution in relation to pond water chemistry in southwestern Ontario, Canada. *Freshwater Biology* 36:7–15.
- Hecnar, S. J. and R. T. M'Closkey. 1997a. The effects of predatory fish on amphibian species richness and distribution. *Biological Conservation* 79:123–131.
- Hecnar, S. J. and R. T. M'Closkey. 1997b. Spatial scale and determination of species status of the green frog (*Rana clamitans melanota*). *Conservation Biology* 11:670–682.
- Hecnar, S. J. and R. T. M'Closkey. 1998. Species richness patterns of amphibians in southwestern Ontario ponds. *Journal of Biogeography* 25:63–772.
- Jeffries, M. 1991. The ecology and conservation value of forestry ponds in Scotland, United Kingdom. *Biological Conservation* 58: 191–211.
- Keddy, P. 1999. Wetland restoration: the potential for assembly rules in the science of conservation. *Wetlands* 19:716–732.
- Kiviat, E. and J. Stapleton. 1983. *Bufo Americanus* (American Toad), estuarine habitat. *Herpetological Review* 14:46.
- Kleinbaum, D. 1994. Logistic Regression: a Self-Learning Text. Springer-Verlag, New York, NY, USA.
- Klemens, M. W., E. Kiviat, and R. E. Schmidt. 1987. Distribution of the northern leopard frog, *Rana pipiens*, in the lower Hudson and Housatonic river valleys. *Northeastern Environmental Science* 6:99–101.
- Lepage, M., R. Courtois, C. Daigle, and S. Matte. 1997. Surveying calling anurans in Quebec using volunteers. *Herpetological Conservation* 1:141–148.
- McAlpine, D. F. and T. G. Dilworth. 1989. Microhabitat and prey size among three species of *Rana* (Anura: Ranidae) sympatric in eastern Canada. *Canadian Journal of Zoology* 67:2244–2252.
- McAlpine, D. F., T. J. Fletcher, S. W. Gorham, and I. T. Gorham. 1991. Distribution and habitat of the Tetraploid Gray Treefrog, *Hyla versicolor*, in New Brunswick and eastern Maine. *Canadian Field Naturalist* 105:526–529.
- McAlpine, D. F. 1997. Historical evidence does not suggest New Brunswick amphibians have declined. *Herpetological Conservation* 1:117–127.
- Ouellet, M. J., J. Bonin, J. Rodrigue, J. Desgranges, and S. Lair. 1997. Hindlimb deformities (ectromelia, ectrodactyly) in free-living anurans from agricultural habitats. *Journal of Wildlife Diseases* 33:95–104.
- Pechmann, J. H. K., D. E. Scott, J. W. Gibbons, and R. D. Semlitsch. 1989. Influence of wetland hydroperiod on diversity and abundance of metamorphosing juvenile amphibians. *Wetland Ecology and Management* 1:3–11.
- Pechmann, J. H. K., D. E. Scott, R. D. Semlitsch, J. P. Caldwell, L. J. Vitt, and J. W. Gibbons. 1991. Declining amphibian populations: the problem of separating human impacts from natural fluctuations. *Science* 253:892–894.
- Perison, D., J. Phelps, C. Pavel, and R. Kellison. 1997. The effects of timber harvest in a South Carolina blackwater bottomland. *Forest Ecology and Management* 90:171–185.
- Pulliam, R. H. 1988. Sources, sinks, and population regulation. *American Naturalist* 132:652–661.
- Rotenberry, J. T. and J. A. Wiens. 1980. Habitat structure, patchiness, and avian communities in North American steppe vegetation: a multivariate analysis. *Ecology* 61:1228–1250.
- Rowe, J. S. 1972. Forest regions of Canada. Canadian Forestry Service, Ottawa, ON, Canada. No. 1300.
- Semlitsch, R. D. and J. R. Bodie. 1998. Are small, isolated wetlands expendable? *Conservation Biology* 12:1129–1133.
- Shirose, L. J., C. A. Bishop, D. M. Green, C. J. MacDonald, R. J. Brooks, and N. J. Helferty. 1997. Validation tests of an amphibian call count survey technique in Ontario, Canada. *Herpetologica* 53: 312–320.
- Sjögren-Gulve, P. 1994. Distribution and extinction patterns within a northern metapopulation of the pool frog, *Rana lessonae*. *Ecology* 75:1357–1367.
- Stevens, C. E., T. S. Gabor, and A. W. Diamond. 2003. Use of restored small wetlands by breeding waterfowl in Prince Edward Island, Canada. *Restoration Ecology* 11:(in press).
- Van Horne, B. 1983. Density as a misleading indicator of habitat quality. *Journal of Wildlife Management* 47:893–901.
- Wake, D. B. 1991. Declining amphibian populations. *Science* 253: 860.
- Warwick, T. 1949. The colonization of bomb-crater ponds at Marlow, Buckinghamshire. *Journal of Animal Ecology* 18:137–141.
- World Meteorological Organization. 1970. The Beaufort Scale of wind force. WMO Commission for Maritime Meteorology, Marine Sciences Affairs Report 3:1–22.
- Zar, J. H. 1999. Biostatistical Analysis, fourth edition. Prentice Hall. Upper Saddle River, New Jersey, USA.

Manuscript received 10 October 2000; revisions received 30 March 2001; accepted 13 November 2001.